UNIVERSITI MALAYSIA SARAWAK | FACULTY OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

Premier Digital Tech Institution

Assignment/Tutorial/Practical Report Cover Sheet
Assignment/Report Cover Sheet

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NUR FATIMA DAMIA BINTI MOHAMMAD YAZIS	107468	

Subject Code:	TMT2713

Project Title:	ART Mobile App Prototype Usability Testing

Due Date: 5 JANUARY 2026	Date Submitted:	4 JANUARY 2026

This cover sheet must be completed, signed and firmly attached to the front of the submission. All work must be submitted by the due date. If an extension of work is granted, an assignment extension acknowledgement slip must be signed by the lecturer/tutor and attached to assignment. Please note that is your responsibility to retain copies of your assignment.

Plagiarism and Collusion are methods of cheating that falls under Peraturan Akademik Universiti Malaysia Sarawak para 11: Etika Akademik

Plagiarism
Plagiarism is the presentation of work which has been copied in whole or in part from another person's work, or from any other source such as the internet, published books or periodicals without due acknowledgement given in the text.

Collusion
Collusion is the presentation of work that is the result in whole or in part of unauthorized collaboration with another person or persons.

Where there are reasonable grounds for believing that cheating has occurred, the only action that may be taken when plagiarism or collusion is detected is for the staff member not to mark the item of work and to report or refer the matter to the Dean. This may result in work being disallowed and given a fail grade or if the circumstances warrant, the matter may be referred to a Committee of inquiry for investigation. Such investigation may result in the matter being referred to the University Discipline Committee, **which** has the power to exclude a student.

Upon placing signature above, I certify that I have not plagiarized the work of others or participated in unauthorized collusion when preparing this assignment.

I also certify that I have taken proper case in safeguarding my work and have made all reasonable efforts to ensure that may work not be able to be copied.

MARK :

Usability Testing Report

Think Aloud Participant Analysis:

User Demographic:

1. Elderly Users:

Our elderly participants generally had limited experience with mobile booking systems. Hence, their feedback focused more on readability, color contrast, font size, and ease of interaction, highlighting accessibility concerns.

2. Working Adults:

The working adults are familiar with mobile apps and valued efficiency and practical features. Their feedback focused on real-world usage scenarios, such as managing bookings and receiving travel reminders.

3. University Students:

Our university students were digitally literate and quick to identify design inconsistencies, incomplete features, and deviations from standard UX patterns found in commercial apps.

Task Overview:

During the usability testing, participants were asked to complete several key tasks, including navigating the booking flow, selecting stations and passengers, confirming a booking, and viewing their booking confirmation and history. Most participants were able to complete the main booking process without major difficulty, and many described the system as simple and straightforward to use. However, some usability issues affected how confidently and efficiently users completed the tasks. Elderly participants, in particular, struggled with small font sizes, low text contrast, and buttons that were placed too close together, which made interaction more challenging. In contrast, working adults and university students completed the tasks more quickly but raised concerns about missing features such as the ability to edit or cancel bookings. These limitations made the system feel incomplete and less practical for real-life use.

Think Aloud Findings and Recommendations:

Demographic	Think-Aloud Feedback	Identified Issue	Design Recommendation
Elderly Users	“The text is hard to see, the colours don’t stand out clearly.”	Low text contrast and thin font weight reduce readability.	Improve color contrast and increase default font sizes to meet accessibility standards.
Elderly Users	“The buttons are very close, I’m scared I’ll press the wrong one.”	Touch targets are too small and closely spaced, increasing error risk.	Increase spacing and size of frequently used buttons.
Working Adults	“The flow is okay, but there’s no option to cancel or edit a booking.”	Missing essential booking management features.	Add edit and cancel booking functionality to support real-world usage.
Working Adults	“It would be helpful if the app reminded me when my ART is arriving.”	Lack of notification or reminder system.	Introduce ART arrival notifications or alarm reminders.
University Students	“The booking system feels incomplete without edit and cancel pages.”	Functional gaps reduce perceived system completeness.	Implement full booking management flows, including edit and cancel options.
University Students	“Some colours don’t match the overall theme, it looks inconsistent.”	Inconsistent color usage disrupts visual consistency.	Standardise fonts and colors across all pages to maintain a coherent theme.
University Students	“There were some spelling errors. For example, in the confirmation page, the word “generated” is misspelled as “genereted”.”	Spelling errors and unclear information grouping reduce trust and credibility.	Fix textual errors and improve information grouping to enhance system credibility.

System Usability Scale Findings

Respondent	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Total	Normalized Total (* 2.5)
1	1	2	3	3	2	2	2	3	2	3	23	57.5
2	3	1	3	1	4	1	4	0	4	1	22	55
3	4	3	4	4	4	4	4	4	4	2	37	92.5
4	3	2	4	1	4	1	3	2	3	1	24	60
5	3	3	3	4	3	1	4	3	2	4	30	75
6	3	3	3	3	2	3	2	3	2	3	27	67.5
7	2	3	3	3	3	2	2	3	3	3	27	67.5
8	2	3	2	1	2	2	1	2	2	1	18	45
9	3	1	3	2	4	2	4	2	3	1	25	62.5
10	2	1	2	1	2	3	3	2	2	0	18	45
11	3	1	2	2	3	2	3	2	3	0	21	52.5
Average:											24.73	61.82

After collective calculation for the finalized score, our prototype got a total score of 61.82. This indicates that our prototype appears complex in terms of the user interface (UI) and the overall user flow is confusing for certain users. This score also indicates that there are inconsistencies in the user interface design which causes some users to be scared to press certain buttons as they are not sure where the buttons would lead them. In conclusion, our application's prototype is currently considered as "Marginal". While the prototype can complete major tasks such as booking, creating an account, login, logout, and reset password, it still lacks the user-friendly interface which is one of the causes of confusion among users.

In order to improve the application, the design of the user interface should be changed to ensure user-friendliness to avoid frustration and confusion among users. A user interface should have consistency in its design which is important as it helps users to detect the similarity of the navigation between different screens. The user flow should also be improved to help navigate the user through each function or process in a smoother flow to ensure users understand the reasoning behind their actions when using the application.

Human Computer Interactions

Project: ART Mobile App Prototype

G02 TMT2713 - Nasi Lemak Bamboo

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Project Overview

Objective:

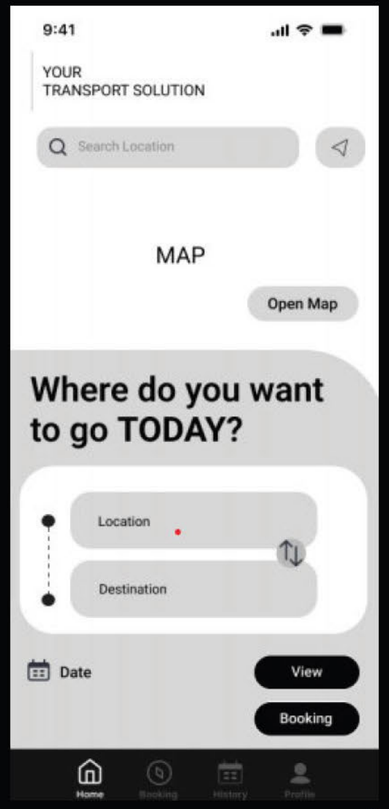
To design a mobile application solution for the ART (Autonomous Rapid Transit) system, focusing on seamless trip planning and ticket booking.

Key Features:

- Secure Login/Sign-up(Email,Phone,Socials).
- Real-time ART Route mapping and station tracking.
- Digital booking and QR code ticket generation.

Initial wireframes (The Design Process)

- Few examples of our wireframe designs



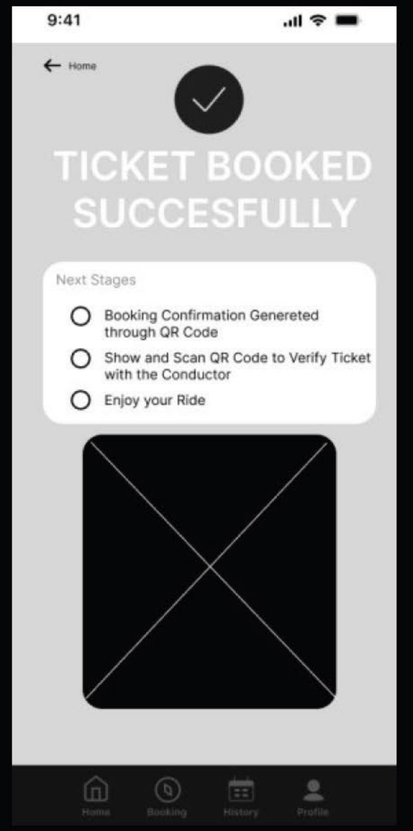
Home page

Show the trip planning interface and search bar.



ART Route Page

Show the live map and station arrival times.



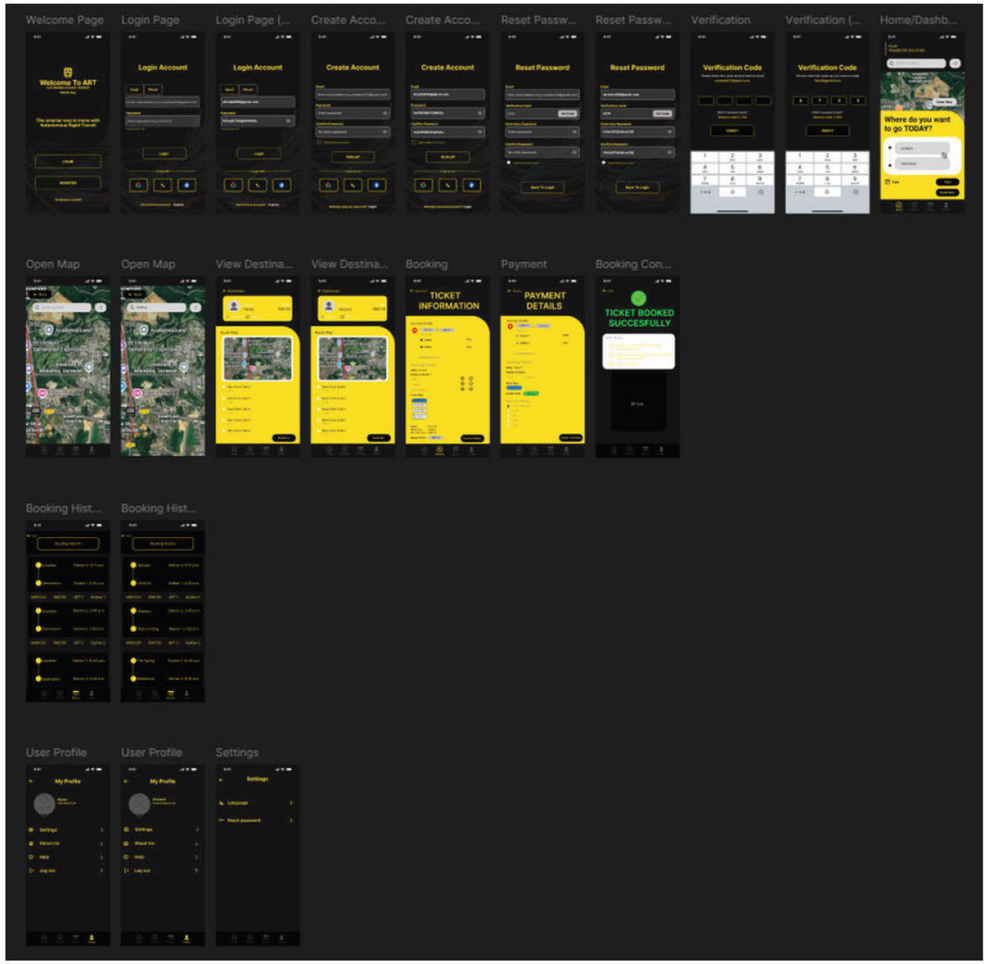
Booking Confirmed Page

Show the final QR code ticket.

Final Prototype

- Transitioning from low-fidelity wireframes to an interactive prototype.
- A smoother flow was established to ensure users understand the reasoning behind their actions, from account creation to booking history.
- FInalised prototype can be seen more clearly here: https://www.figma.com/design/VlVdR3oWp2HyRJytwNHmBn?node-id=0-1

Overview of the prototype



Usability Testing Methodology

- **Participants**
  11 Respondents

- **Demographics**
  - Elderly Users: Focused on accessibility and readability.
  - Working Adults: Focused on efficiency and real-world utility.
  - University Students: Focused on UX patterns and design consistency.
  - Tasks: Navigating booking flows, selecting stations, and viewing booking history.

Think Aloud Findings

- **Accessibility (Elderly):**
  Low text contrast and small touch targets made interaction difficult.
- **Functionality (Adults/Students):**
  Users noted missing "Edit" and "Cancel" features for bookings.
- **UI Consistency (Students):**
  Identified spelling errors (e.g., "genereted") and inconsistent color themes.

System Usability Scale (SUS) Results

Average Score: 61.82

Grade: “Marginal”

Analysis:

1. The UI appeared complex to some users.

2. Inconsistencies caused hesitation when pressing buttons.

3. Major tasks (booking/login) were successful, but user-friendliness needs improvement.

Design Recommendations & Next Steps

Improve Accessibility:
Increase font sizes and color contrast; enlarge touch targets.

Visual Standardisation:
Implement full booking management (Edit/Cancel) and arrival notifications.

Feature Expansion:
Fix textual errors and standardize fonts/colors across all screens to improve credibility.

THANK YOU!

Usability Testing Consent Form

Project: Mobile Application Prototype for Kuching Urban Transportation System (ART)

Institution: Universiti Malaysia Sarawak (UNIMAS)

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- Complete a short survey (System Usability Scale) at the end of the session to rate your experience. (Only applicable to those who are participating in System Usability Scale test)

Terms of Participation

Voluntary: Your participation is entirely voluntary. You may withdraw at any time without penalty.

Recording: To help us analyze the data, we will record your screen interactions and voice (audio). These recordings are for internal analysis only.

Confidentiality: Your participation is strictly confidential. Your name will not appear in the final report; you will be referred to by a generic identifier (e.g., "Participant 1"). If you have questions about this study, you may contact:

Name: Darryiell Karllen

Email: 103793@siswa.unimas.my

By signing below (or checking the box), I acknowledge that:

[✓] I have read and understood the information above.

[✓] I agree to participate in this usability study.

[✓] I agree to have my voice and screen interactions recorded for analysis.

Participant Name: [redacted]

Signature: [signature]

Date: 31/01/2026

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Date: 3 Jan 2026

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Participant Name: [redacted]

Signature: _____

Date: 24/12/2025

Usability Testing Consent Form

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Participant Name: [redacted]

Signature: Cornelia

Date: 31 December 2025

Usability Testing Consent Form

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Participant Name: [redacted]

Signature: _____

Date: 25 December 2025

Usability Testing Consent Form

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Participant Name: [redacted]

Signature: _____

Date: 25 DECEMBER 2025

System Usability Scale Form

SUS Questionnaire	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. I think I would like to use this system frequently.	1	(2)	3	4	5
2. I found this system unnecessarily complex.	1	2	(3)	4	5
3. I thought the system was easy to use.	1	2	3	(4)	5
4. I think that I would need the support of a technical person to be able to use this system.	1	(2)	3	4	5
5. I found the various functions in this system were well integrated.	1	2	(3)	4	5
6. I thought there were too much inconsistency in this system.	1	2	(3)	4	5
7. I would imagine that most people would learn to use this system very quickly.	1	2	(3)	4	5
8. I found the system very cumbersome to use.	1	(2)	3	4	5
9. I felt very confident using the system	1	2	(3)	4	5
10. I needed to learn a lot of things before I could get going with the system.	1	(2)	3	4	5

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Signature: _____

Date: 03 Jan 2026

System Usability Scale Form

SUS Questionnaire	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. I think I would like to use this system frequently.	1	2	3	(4)	5
2. I found this system unnecessarily complex.	1	2	3	(4)	5
3. I thought the system was easy to use.	1	2	3	(4)	5
4. I think that I would need the support of a technical person to be able to use this system.	1	2	3	(4)	5
5. I found the various functions in this system were well integrated.	1	2	3	4	(5)
6. I thought there were too much inconsistency in this system.	1	2	3	(4)	5
7. I would imagine that most people would learn to use this system very quickly.	1	2	3	4	(5)
8. I found the system very cumbersome to use.	1	2	3	4	(5)
9. I felt very confident using the system	1	2	3	4	(5)
10. I needed to learn a lot of things before I could get going with the system.	1	2	3	(4)	5

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Date: 03 / 01 / 2026

System Usability Scale Form

SUS Questionnaire	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. I think I would like to use this system frequently.	1	2	3	4	5
2. I found this system unnecessarily complex.	1	2	3	4	5
3. I thought the system was easy to use.	1	2	3	4	5
4. I think that I would need the support of a technical person to be able to use this system.	1	2	3	4	5
5. I found the various functions in this system were well integrated.	1	2	3	4	5
6. I thought there were too much inconsistency in this system.	1	2	3	4	5
7. I would imagine that most people would learn to use this system very quickly.	1	2	3	4	5
8. I found the system very cumbersome to use.	1	2	3	4	5
9. I felt very confident using the system	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with the system.	1	2	3	4	5

Usability Testing Consent Form

Project: Mobile Application Prototype for Kuching Urban Transportation System (ART)

Institution: Universiti Malaysia Sarawak (UNIMAS)

Introduction & Purpose: You are invited to participate in a usability testing session for a new mobile application prototype designed for the Autonomous Rapid Transit (ART) system in Kuching and Samarahan. The purpose of this study is to evaluate the ease of use and functionality of the app design. Below are the procedures If you agree to participate, the session will take approximately **15 to 20 minutes.** You will be asked to:

- Perform specific tasks using the mobile app prototype (e.g., planning a route, purchasing a ticket, checking schedules).

- "Think Aloud" while performing these tasks. This means speaking your thoughts, confusions, and decisions out loud as you navigate the app.

- Complete a short survey (System Usability Scale) at the end of the session to rate your experience. (Only applicable to those who are participating in System Usability Scale test)

Terms of Participation

Voluntary: Your participation is entirely voluntary. You may withdraw at any time without penalty.

Recording: To help us analyze the data, we will record your screen interactions and voice (audio). These recordings are for internal analysis only.

Confidentiality: Your participation is strictly confidential. Your name will not appear in the final report; you will be referred to by a generic identifier (e.g., "Participant 1"). If you have questions about this study, you may contact:

Name: Darryiell Karllen

Email: 103793@siswa.unimas.my

By signing below (or checking the box), I acknowledge that:

[ ] I have read and understood the information above.

[ ] I agree to participate in this usability study.

[ ] I agree to have my voice and screen interactions recorded for analysis.

Participant Name: [redacted]

Signature: M

Date: 3/1/26

System Usability Scale Form

SUS Questionnaire	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. I think I would like to use this system frequently.	1	2	3	(4)	5
2. I found this system unnecessarily complex.	1	2	(3)	4	5
3. I thought the system was easy to use.	1	2	3	4	(5)
4. I think that I would need the support of a technical person to be able to use this system.	1	2	3	(4)	5
5. I found the various functions in this system were well integrated.	1	2	3	4	(5)
6. I thought there were too much inconsistency in this system.	1	2	3	(4)	5
7. I would imagine that most people would learn to use this system very quickly.	1	2	3	(4)	5
8. I found the system very cumbersome to use.	1	2	(3)	4	5
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[/] I agree to participate in this usability study.

[/] I agree to have my voice and screen interactions recorded for analysis.

Participant Name: Hazirah

Signature: 

Date: 25/12/2025

System Usability Scale Form

SUS Questionnaire	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. I think I would like to use this system frequently.	1	2	3	(4)	5
2. I found this system unnecessarily complex.	1	(2)	3	4	5
3. I thought the system was easy to use.	1	2	3	(4)	5
4. I think that I would need the support of a technical person to be able to use this system.	(1)	2	3	4	5
5. I found the various functions in this system were well integrated.	1	2	3	(4)	5
6. I thought there were too much inconsistency in this system.	1	2	3	(4)	5
7. I would imagine that most people would learn to use this system very quickly.	1	2	3	4	(5)
8. I found the system very cumbersome to use.	1	(2)	3	4	5
9. I felt very confident using the system	1	2	(3)	4	5
10. I needed to learn a lot of things before I could get going with the system.	(1)	2	3	4	5

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Participant Name: [redacted]

Signature: _____

Date: 26 DECEMBER 2025

System Usability Scale Form

SUS Questionnaire	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. I think I would like to use this system frequently.	1	2	3	(4)	5
2. I found this system unnecessarily complex.	1	(2)	3	4	5
3. I thought the system was easy to use.	1	2	3	(4)	5
4. I think that I would need the support of a technical person to be able to use this system.	1	(2)	3	4	5
5. I found the various functions in this system were well integrated.	1	2	(3)	4	5
6. I thought there were too much inconsistency in this system.	1	(2)	3	4	5
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9. I felt very confident using the system	1	2	(3)	4	5
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[✓] I agree to have my voice and screen interactions recorded for analysis.

Participant Name: [redacted]

Signature: [signature]

Date: 31/12/2025

System Usability Scale Form

SUS Questionnaire	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. I think I would like to use this system frequently.	1	2	(3)	4	5
2. I found this system unnecessarily complex.	1	(2)	3	4	5
3. I thought the system was easy to use.	1	2	3	(4)	5
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Participant Name [redacted]

Signature: _____

Date: 25 DECEMBER 2025

System Usability Scale Form

SUS Questionnaire	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1. I think I would like to use this system frequently.	1	2	(3)	4	5
2. I found this system unnecessarily complex.	1	2	(3)	4	5
3. I thought the system was easy to use.	1	2	(3)	4	5
4. I think that I would need the support of a technical person to be able to use this system.	1	2	3	(4)	5
5. I found the various functions in this system were well integrated.	1	2	(3)	4	5
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Date: 03/01/2026

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2. I found this system unnecessarily complex.	1	2	3	(4)	5
3. I thought the system was easy to use.	1	2	3	4	(5)
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